College of …… Information Engineering
Department… Software

الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

Car Rental and Sale Platform Project

Student Preparation

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Supervision

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Importance of the project:

The project's expected value lies in providing a secure and smart online platform that shortens the distances between car service providers and car seekers. The importance is highlighted by the following:

1. Raise the level of transparency: by displaying accurate details, images, and comprehensive specifications for each car.
2. Save time and effort: Aggregate buying and leasing options in one place with advanced search and filtering tools.
3. Organizing business processes: Automate the cycle of booking cars and display ads in a way that reduces human errors and ensures a smooth and professional user experience

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Project Objectives :

The project aims to achieve the following in a practical, measurable and verifiable manner:

1 - Designing and developing an integrated web system to manage the sales, purchase, and rental of cars (new and used) with high efficiency

2 - Building a flexible and modern user interface using (Tailwind CSS and Blade) that ensures a comfortable and responsive browsing experience for various devices.

3 - Providing an advanced admin panel for user management, ad review, car rating, and permission control.

4- Securing data and organizing software relationships using the MySQL database and the framework (Laravel 13) to ensure speed, reliability and security of transactions .

Reference Studies and Previous Works

Study I: Building a Computer Platform for Managing Car Showrooms Using Basic Web Technologies and MySQL Databases

Study Two: A Comparison of Modern Web Frameworks in the Development of High-Load E-Commerce Systems

Study Three: The Impact of Adaptive Interface Design Using Tailwind CSS on User Experience in Fast Rental Platforms for Vehicles

Fourth Study: ..... [....]

Fifth Study: ..... [....]

Study Six: ..... [....]

Comparison of previous studies

Strengths	Techniques used	Feature2	Feature1	Study
Simple automation of vehicle data and information entry .	Responsiveness is slow and limited with traditional screene	Sell cars only	Native PHP, MySQL, Bootstrap	. **First Study [1] (2024)**
Deliver a secure and reliable infrastructure that adopts the MVC architecture and prevents vulnerabilities	Good performance with safe order processing	Single management system for sales services	Laravel, MySQL, REST API	**Study II [2] (2025)**
Significantly increase the speed of platform responsiveness and the responsiveness of different screens	Responsive and ultra-lightweight interfaces	Car rental service only	Vue.js, Node.js, Tailwind CSS	**Study III [3] (2026)**

General Functional System Requirements

(Code) Symbol)	General Career Requirement
FR-G01	The system should provide an interactive environment for user registration and permission management for the various system roles
FR-G02	Enable visitors and customers to search for and browse cars with high flexibility
FR-G03	Tools to efficiently manage car ads, reservation and purchase requests
FR-G04	Providing administrative and personal control interfaces to monitor and organize operations

Non-Functional Requirements

Explanation	Non-Functional Requirement	Code(s)
Ensure server responsiveness and load web pages as quickly as possible thanks to optimized MySQL database queries .	Performance and responsiveness	**NFR-01**
Secure the system against common vulnerabilities such as SQL injection, order forgery (CSRF), and password encryption with .Laravel's built-in features	Security and protection	**NFR-02**
The user interfaces are fully compatible with various screen sizes smart devices using the Tailwind CSS design and framework.	Responsiveness and flexibility	**NFR-03**
Adopting an MVC architecture and an organized structure that allows modifying and developing code and adding new features in the .future with ease	Maintainability and Scalability	**NFR-04**

Job Requirements for the Semester Project

Priority	Symbol	Requirement	Description
١	FR-01	Authentication and Permission Management	A secure system for creating accounts and distributing permissions (visitor, customer, advertiser, manager).
٢	**FR02-**	Advanced Search & Filter	Accurate search engine and filter of cars by type, brand, price, and year of manufacture
٣	**FR-03**	Manage car ads	Enable advertisers to perform CRUDs to add, edit, and .delete car ads
٤	**FR-04**	Car Reservation & Rental	Enable customers to set dates for booking cars for rental and calculate their time cost .
٥	**FR-05**	Purchase Orders & Inquiry	Send direct purchase orders and official inquiries about the cars for sale.
٦	**FR-06**	Administrative Control Panel	The administrator's authority to review, accept/reject advertisements and manage the entire system users.
٧	**FR-07**	User and Advertiser Dashboard	A personalized interface to keep track of the status of published announcements and incoming and sent requests.
٨	**FR-08**	Vehicle Details & View	An integrated detailed page for each car showcasing its technical specifications and interactive photo gallery.
٩	**FR-09**	Notifications & Alerts	An internal alert system that notifies users of any updates regarding the status of their orders or ads.
١٠	**FR-10**	Ratings & Reviews	It allows customers to write ratings and reviews about the level of service and the credibility of advertisers.

Development Methodology:

The Agile Development Methodology and specifically the Scrum framework have been adopted in the development of a car rental and sales platform

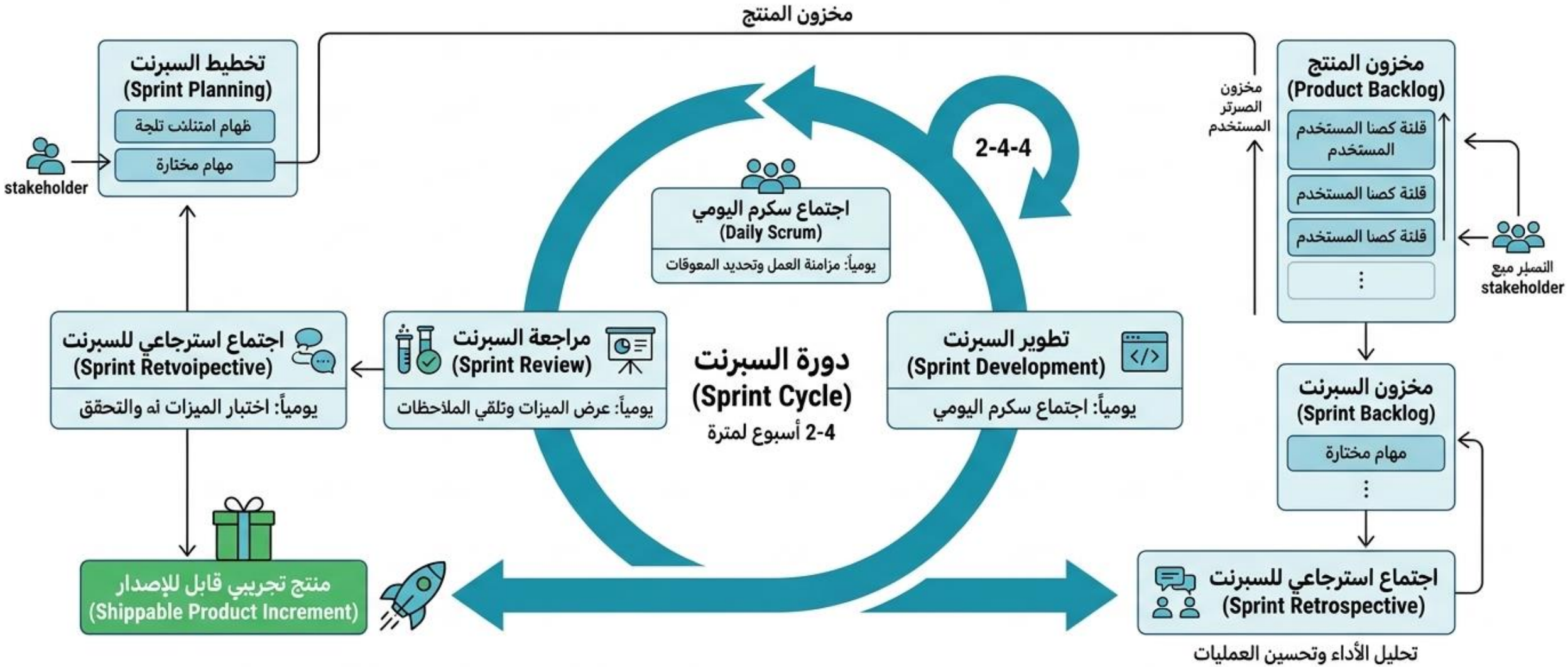
This methodology has been adopted because: it is highly flexible in dealing with changing requirements, and allows the project to be segmented into short work cycles (sprints), which facilitates the process of evaluating software outputs first-hand, detecting and addressing errors early, and ensuring that the system is gradually delivered with high efficiency and in line with the needs of users

Project Chart Gantt Time Plan (Example)

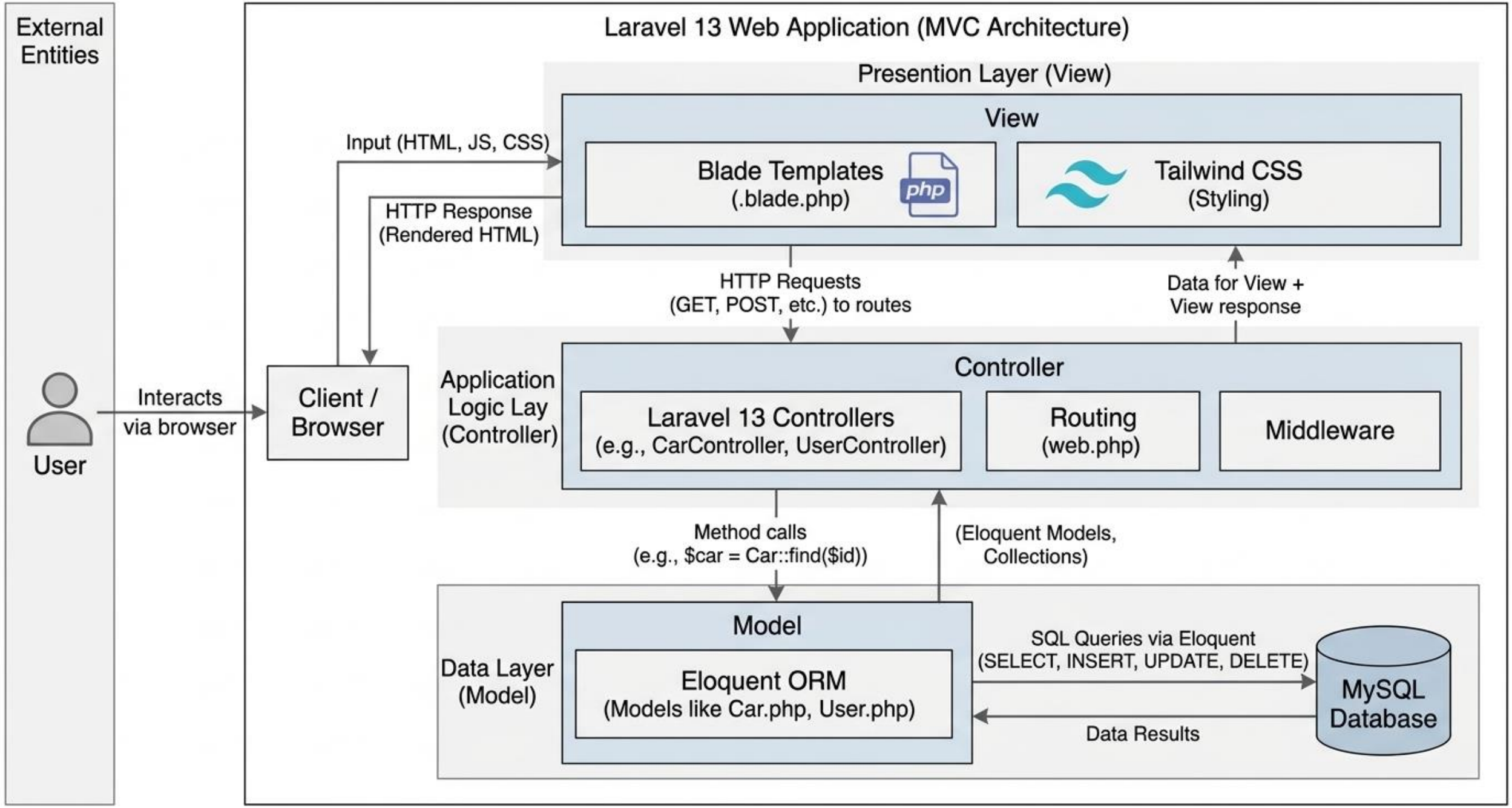


منهجية أجايل سكْرم (Agile Scrum Methodology)

منصة بيع وتأجير السيارات



مثل: مخرج، معام، عملية، مراجعة


External Entities
Laravel 13 Web Application (MVC Architecture)
Presention Layer (View)
View
Input (HTML, JS, CSS)
Blade Templates
(.blade.php)
php
Tailwind CSS
(Styling)
HTTP Response
(Rendered HTML)
HTTP Requests
(GET, POST, etc.) to routes
Data for View +
View response
User
Interacts
via browser
Client /
Browser
Application
Logic Lay
(Controller)
Controller
Laravel 13 Controllers
(e.g., CarController, UserController)
Routing
(web.php)
Middleware
Method calls
(e.g., $car = Car::find($id))
(Eloquent Models,
Collections)
Data Layer
(Model)
Model
Eloquent ORM
(Models like Car.php, User.php)
SQL Queries via Eloquent
(SELECT, INSERT, UPDATE, DELETE)
Data Results
MySQL
Database